Describe the applicable skills and abilities you have gained through relevant research and work experience.

Personal and Technical Profile

I have built a strong foundation in computer systems, networking, and software development through my degree in Management Information Systems, complemented by hands-on technical experience and research involvement.

My education has strengthened my analytical and problem-solving skills, while my ongoing preparation for CCNA certification has deepened my understanding of network infrastructure, IoT systems, and network security. In parallel, I have been enhancing my front-end web development proficiency using HTML, CSS, JavaScript, and React. Through small-scale projects, I have gained valuable experience in designing responsive, functional, and user-friendly interfaces - skills that enable me to create efficient and visually appealing web applications.

I am also developing my skills in Python, focusing on its applications in back-end development, automation, and data processing. Although still in the learning phase, I continually strengthen my abilities through consistent practice and personal projects. This journey has nurtured my perseverance, curiosity, and commitment to lifelong learning—qualities essential for growth and innovation in the technology sector.

My professional experience has further refined my leadership, teamwork, and communication skills. Collaborating with diverse teams in dynamic environments has taught me to remain adaptable, think critically, and approach challenges with focus and creativity.

Collectively, these experiences have shaped me into a dedicated and versatile learner with a strong passion for technology and innovation. I am confident that this scholarship will provide the platform to deepen my technical expertise and enable me to make meaningful contributions to technological advancement within my community and beyond.

LAY ABSTRACT (Please describe your research project in **LAY TERMS** within **100 words or less**)

MIVA-KNIGHT is a voice-driven AI assistant that helps industries make faster, smarter, and safer decisions by analyzing speech, text, images, and sensor data in real time. Combining knowledge graphs, hybrid RAG-LightRAG retrieval, and explainable AI, it delivers actionable insights on site while keeping data fully private. Designed for healthcare, energy, manufacturing, finance, and education, MIVA - KNIGHT reduces downtime, boosts accuracy, and cuts operational costs - offering a turnkey, scalable solution for next-generation industrial AI.

PART 1 A: Outline of Research (maximum one page) Please read instructions before completing this section.

Novelty Statement : MIVA - KNIGHT introduces the first industrial-grade, end-to-end multi-modal voice assistant integrating knowledge graphs, hybrid RAG - LightRAG retrieval, and explainable AI (XAI) for real-time, auditable reasoning across text, images, voice, and sensor data [1 - 4]. Unlike existing multi-modal LLMs [5 - 7], it combines structured domain knowledge and multi-modal understanding for healthcare, energy, manufacturing, finance, and education, ensuring data sovereignty and interpretable decision-making [8 - 10].

Background and Problem Relevance: Modern industries must interpret vast, diverse data - manuals, sensor signals, medical images, and reports - under time pressure. While LLMs like GPT and BERT exhibit strong reasoning [6, 24], they lack structured domain integration [4, 26], cross-modal understanding [1, 2, 11], and on-premise interpretability [17 - 19]. These gaps hinder adoption in privacy-critical fields such as healthcare and energy [20 - 22]. Alberta's diversified economy urgently needs a voice-driven AI assistant combining expert reasoning, conversational interaction, and secure deployment.

Scientific and Technological Innovation: MIVA - KNIGHT fuses RAG-LightRAG hybrid retrieval, multi-modal transformers, and visual XAI in one reasoning framework [3 - 5,11,17]. Key contributions include: i) Hybrid RAG-LightRAG: Dense embedding retrieval plus graph-based multi-hop reasoning [3, 4, 26]. ii) Explainable Reasoning: Generates attention maps and reasoning trails [17 - 19]. iii) End-to-End On-Premise Deployment: Containerized microservices for privacy and real-time operation [20, 21]. iv) This unified system enables cross-modal understanding and human-verifiable decisions across industrial domains.

Objectives: A. Domain Knowledge Graphs: Build healthcare, energy, manufacturing, finance, and education graphs (10k–50k entities/domain) [26 - 27]. Milestone: Month 4.
B. Hybrid Retrieval: Implement RAG-LightRAG with ≥15% accuracy gain vs. baseline and ≥25% over graph-only models [3 - 5].
C. Multi-modal Understanding: Align text, images, voice, and sensor data (cosine similarity ≥0.8) [1, 2,11,14].

D. Explainable Reasoning: Achieve ≥90% expert-rated interpretability [17 - 19].

E. Real-Time Operation: Sub-2 s latency, on-premise [20 - 22].

Target Impacts: Healthcare: +30% accuracy, - 80% diagnosis time [20]; Energy: - 15% cost [22,29]; Manufacturing: - 40% downtime [21]; Finance: +25% risk accuracy; Education: +50% engagement, - 60% search time [25].

Methodology : Phase 1 (Months 1 - 4): Graphs & Datasets Develop Neo4j-based graphs [26–27]. Collect multimodal data: Voice/Audio: Multi-modal Pathological Voice Classification, RAVDESS, CREMA-D, MELD, Medical Speech and Intent (Kaggle/GitHub) [11 - 15]. Text: Academic, industrial, and institutional documents [24, 25]. Images/Videos: Annotated healthcare, industrial, and educational data [8, 21, 23]. Train baselines: BERT (text) [24], ResNet (image) [23], Whisper (speech) [11], LSTM (time-series).

Phase 2 (Months 5–8): Hybrid Retrieval & Fusion Implement RAG-LightRAG (vector + graph retrieval) [3, 4, 26]. Train multi-modal transformers for unified embeddings [1, 2, 8]. Develop predictive modules: U-Net (medical), energy forecasting [22, 29], and educational retrieval [25].

Phase 3 (Months 9 - 10): Explainable Interface Integrate Whisper ASR and neural TTS [11,15]. Implement reasoning-path visualization and graph attention maps [17 - 19].

Phase 4 (Months 11 - 12): Evaluation & Deployment Assess accuracy, latency, interpretability, and user satisfaction [20 - 22]. Deploy via Docker for secure, on-premise rollout.

Expected Outcomes: Fully functional MIVA-KNIGHT prototype in five domains. Domain graphs (10k–50k entities each) [26 - 27]. ≥85% accuracy, <2 s latency, ≥90% interpretability. 1 - 2 publications in reputed conferences or journals [1 - 7,17,21]. Open-source, containerized framework for multi-modal AI research.

Scientific Merit: MIVA - KNIGHT advances multi-modal reasoning, hybrid retrieval, and XAI [1 - 5,17–19,30]. It transitions AI from unimodal perception to knowledge-grounded, multi-modal understanding within privacy-preserving, explainable architectures. This project will deliver tangible, domain-specific impact and position Alberta as a leader in secure, human-centric AI innovation.

PART 1 B: Bibliography/Citations (maximum one page) Please read instructions before completing this section.

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PART 2: Alberta Innovates Research Alignment - IMPACT FOR ALBERTA (maximum one page) Please read instructions before completing this section.

Clearly articulate in the application how the results of the research can produce potential environmental, health, economic, social and other impacts for Alberta. Address the potential for how your technology/proposed solution is to be adopted in Alberta and beyond. **Note: The assessment criteria and all investment decisions are at the sole discretion of Alberta Innovates.**

1. Environmental and Energy Impact: Alberta's economy remains heavily reliant on energy-intensive sectors that account for over 38% of provincial GHG emissions (Environment and Climate Change Canada, 2023). MIVA-KNIGHT enables energy operators to process multi-modal data - sensor readings, maintenance logs, and audio reports - to identify inefficiencies in real time. By fusing knowledge graphs and hybrid RAG-LightRAG retrieval, the system can recommend optimized load management, predictive maintenance schedules, and fault diagnosis, yielding 15 - 20% energy savings (NRCan, 2020) and directly contributing to Alberta's net-zero 2050 target. Explainable AI ensures every recommendation is transparent and auditable, meeting emerging ISO/IEC AI governance and Alberta Innovates environmental compliance standards.

2. Health and Biomedical Impact: Healthcare systems in Alberta face diagnostic bottlenecks, data silos, and high administrative burden. MIVA-KNIGHT's multi-modal capability - combining medical images, patient voice data, and text records - allows clinicians to query data conversationally and receive interpretable diagnostic insights. By leveraging domain-specific knowledge graphs and XAI, diagnostic turnaround time can be reduced by up to 80%, while interpretability exceeds 90% expert-rated transparency. These capabilities align with Alberta Health's Precision Health Strategy (2022) and can support telemedicine in rural communities where healthcare access is limited. Cleaner decision processes and reduced diagnostic delays indirectly enhance public health by improving treatment response times and reducing clinical error rates.

3. Economic and Industrial Impact: Industrial automation and digital transformation are key to Alberta's diversification agenda. By enabling AI-driven predictive maintenance, MIVA-KNIGHT can reduce unplanned downtime by 30 - 40% and extend equipment lifespan, resulting in significant operational cost savings (McKinsey Global Institute, 2022). In energy-intensive industries such as petrochemicals and heavy manufacturing, even a 1% improvement in process efficiency translates to tens of millions of dollars in annual savings (IEA, 2023). Furthermore, Alberta's AI ecosystem - valued at over $1.4 billion in 2024 (Alberta Innovates Report) - stands to benefit from local development and on-premise deployment of privacy-preserving AI systems, stimulating job creation in data science, software, and energy analytics sectors.

4. Financial and Governance Impact: The finance sector increasingly demands interpretable AI for fraud detection, credit scoring, and compliance monitoring. MIVA-KNIGHT's explainable, graph-integrated reasoning allows for transparent audit trails - a critical compliance requirement under OSFI and GDPR-like AI regulations. Studies from the OECD (2023) show explainable AI can increase decision accuracy by 20–30% and cut audit costs by up to 15%. This project thus strengthens Alberta's growing fin-tech ecosystem by providing locally deployable, explainable models that maintain both accuracy and accountability.

5. Social and Educational Impact: Education and workforce transformation are essential to Alberta's innovation agenda. By introducing voice-driven, multi-modal learning interfaces, MIVA - KNIGHT supports personalized education and energy/AI literacy across schools and technical institutes. OECD studies (2023) report that AI-based adaptive tutoring can increase learner engagement by 50% and reduce information retrieval time by 60%. Public access to interpretable AI systems also enhances AI literacy, fostering a more informed and equitable society. The project supports Alberta Innovates' talent and skills development objectives, ensuring a technically proficient workforce to sustain AI adoption.

6. Broader Adoption and Global Relevance: MIVA-KNIGHT's on-premise, containerized architecture ensures full data sovereignty - a major barrier for AI deployment in healthcare, finance, and energy sectors. This design facilitates seamless adoption by Alberta industries while maintaining compliance with Canadian privacy regulations (PIPEDA, 2023). Its modular design allows global scalability, positioning Alberta as a leader in explainable, multi-modal AI exports, with potential licensing opportunities to European and Asia - Pacific industries undergoing similar digital transitions.

All references cited in this section are provided in Part 3: Refereed Contributions, Commissioned, Juried, and/or Invited Exhibitions (Page 13).